Taqwatech at IslamicEval 2026: Contextual Span Detection and Corpus-Grounded Verification of Islamic Citations

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Abstract

Arabic large language models can generate fluent religious answers but may misquote or misattribute Qur'anic and Hadith evidence. This makes reliable verification important in a domain where the accuracy of the quoted text and its source matters. We describe Taqwatech's systems for Subtasks 1 and 2 of IslamicEval 2026. For Subtask 1, we use CAMeLBERT-Mix with BIO tagging to find Ayah, Hadith *matn*, *isnad*, and claimed-source spans. We then convert the predicted token labels back to the character positions required by the task. For Subtask 2, we use a verification system based on the cited sources that combines Arabic normalization, lexical retrieval, character TF-IDF, Qur'anic verse matching, source checks, and narrator matching. On the official blind test, our systems achieved 0.9683 character-level macro-F1 for Subtask 1 and 0.9050 macro accuracy for Subtask 2. The systems perform well for span detection and Qur'anic verification, while narrator-chain verification remains the main challenge.

1 Introduction

Large language models (LLMs) are widely used for question answering and text generation in Arabic. Although their answers are usually clear and fluent, they may contain information that is incorrect or unsupported. This problem is known as hallucination (Rawte et al., 2023). The problem is particularly important in religious questions, where the accuracy of quotations and their sources is important.

The Qur'an and Hadith are central sources of Islamic knowledge. However, an LLM may change part of a Qur'anic verse, produce an unsupported Hadith, connect a correct quotation to the wrong source, or mention an incorrect narrator. These errors are not always easy to notice because the generated answer may still sound natural. Checking fluency alone is therefore not enough; the quotation must be compared with trusted sources.

IslamicEval 2025 introduced a shared task for detecting and correcting hallucinated Qur'anic and Hadith content in Arabic LLM responses (Mubarak et al., 2025). IslamicEval 2026 extends this work through a more detailed annotation of each citation (Alharbi et al., 2026). A citation may contain an Ayah, a Hadith *matn*, an *isnad*, and a claimed source. This allows each part of a citation to be checked separately.

Taqwatech participated in the first two subtasks. Subtask 1 asks systems to find the exact character positions and types of citation segments. Subtask 2 provides the segments and asks whether each one is correct or incorrect. We use contextual BIO tagging for the first task and source-based verification for the second. On the official blind test, our systems achieved 0.9683 character-level macro-F1 for Subtask 1 and 0.9050 macro accuracy for Subtask 2.

2 Task and Data

Each example contains a user question and an Arabic answer generated by an LLM. An answer can contain no citation, a single citation, or several citations. Each citation is divided into one or more segment types: Ayah, Hadith *matn*, *isnad*, and claimed source.

An Ayah segment contains quoted Qur'anic text. A *matn* contains the main wording of a Hadith. An *isnad* contains narrator information. A claimed source may contain a surah name, verse number, or

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Split	Resp.	Cites	Spans	S2 scored
Train	4,706	15,826	32,779	24,088
Dev	484	1,358	2,728	1,745

Table 1: Statistics of the released data. The final column excludes N/A segments from Subtask 2.

the name of a Hadith collection.

2.1 Subtask 1: Span Detection

In Subtask 1, the system must identify every citation segment in the generated answer. For each segment, the system returns its type and its start and end character positions. Segments belonging to the same citation share an annotation ID. If the answer has no citation, the system returns the required `NoAnnotation` entry.

The official evaluation metric is character-level macro-F1. Every character is assigned to one of five classes: ayah, *matn*, *isnad*, claimed source, or neither. This evaluation measures both the predicted type and the quality of the span boundaries.

2.2 Subtask 2: Hallucination Identification

In Subtask 2, the citation segments and their types are already provided. The system predicts `correct` or `incorrect` for each segment. An ayah is checked against the canonical Qur'an, while a Hadith *matn* is checked against the supplied Hadith records.

Claimed-source and *isnad* segments depend on their main quotations. They are evaluated only when the related Ayah or *matn* is correct. If the main quotation is incorrect, these segments receive the N/A label and are excluded from scoring. Accuracy is calculated for each segment type, and the final score is calculated as the unweighted average of the four accuracy values.

2.3 Dataset and Reference Collections

We used the official training, development, and test files. The training split contains 4,706 responses, 15,826 citations, and 32,779 segments. The development split contains 484 responses, 1,358 citations, and 2,728 segments. After excluding N/A labels, 24,088 training segments and 1,745 development segments are available for Subtask 2.

The organizers also supplied a canonical Qur'an collection and a Hadith collection covering six books. We use these collections as trusted sources for verification. No externally labeled examples are added.

3 Subtask 1 System

3.1 Approaches

We tried two approaches for citation span detection. The first approach treats the task as BIO sequence labeling and fine-tunes CAMeLBERT-Mix, an Arabic pretrained language model (Inoue et al., 2021).. The second approach uses Qwen3-4B (Yang et al., 2025), fine-tuned with QLoRA (Dettmers et al., 2023), to generate a JSON list containing the citation type and its character offsets.

The generative approach could not be evaluated completely so that it could be compared fairly with the sequence-labeling system. We therefore selected CAMeLBERT-Mix for our official submission. This system achieved a character-level macro-F1 score of 0.9683 on the official test set.

3.2 BIO Label Generation

First, we assign a label to each character in the generated answer. The labels include O and beginning/inside labels for each of the four segment types. This results in nine labels in total:

$$\{\text{O}, \text{B/I-Ayah}, \text{B/I-matn}, \text{B/I-isnad}, \text{B/I-source}\}.$$

The CAMeLBERT tokenizer provides an offset mapping between tokens and characters. We use this mapping to transfer the character annotations to token-level BIO labels. Special and padding tokens are ignored during training.

3.3 Model and Training

We fine-tune CAMeLBERT-Mix (Inoue et al., 2021)[1] with a token-classification head. We use this model because the generated answers may combine modern Arabic explanations with classical Qur'anic and Hadith text.

We split the answers into 512-token windows with a stride of 128 tokens. The overlap helps avoid losing citations in long answers before all citations are processed. We use AdamW with a learning rate of 2×10^{-5}, training batch size 8, evaluation batch size 16, and weight decay 0.01. Training is limited to 30 epochs with early stopping after three evaluations without improvement. The random seed is set to 42.

[1] https://huggingface.co/CAMeL-Lab/bert-base-arabic-camelbert-mix

3.4 Span Decoding

During prediction, each token receives a BIO label. We use the stored offset mapping to convert these labels back to character positions. Predictions of the same type are joined when they overlap or touch across two windows.

The task output also requires an annotation ID for each citation. Since the model predicts flat segments, we order them by their starting position and begin a new citation group when the gap from the previous segment is greater than 50 characters. Responses without a predicted citation receive a `NoAnnotation` row.

4 Subtask 2 System

4.1 System Overview

We approach hallucination identification as verification against the provided sources rather than as a general binary classification problem. First, the system normalizes each Arabic fragment, retrieves possible matches from the provided collections, and then applies checks designed for the segment type.

The system consists of several main components:

- Arabic text normalization;
- lexical retrieval for Qur'anic and Hadith text;
- character TF–IDF retrieval for Hadith wording differences;
- memory of known labelled and accepted variants;
- Qur'anic location and Hadith collection checks;
- narrator-chain matching; and
- rules for short, misplaced, or wrongly attributed fragments.

4.2 Arabic Normalization

Arabic text may contain diacritics, tatweel, and different forms of the same letter. These differences can reduce the similarity score even when the text is otherwise the same. We remove diacritics and tatweel, normalize the main alef forms, map alif maqsura to ya, and normalize hamzated waw and ya. We use normalization only for comparison; the original task text and output keys remain unchanged.

4.3 Ayah Verification

The Qur'an index contains individual verses and windows of up to five consecutive verses. These windows are useful when an answer quotes text that crosses a verse boundary.

Candidates are first found using word-based retrieval. They are then compared using character-gram containment, which allows a short quotation to match a longer canonical passage. An Ayah is accepted when the best score reaches 0.970.

The system also checks whether an exact Qur'anic quotation appears at the claimed location. This prevents the system from accepting a correct verse from being accepted when it is linked to a different surah or verse number.

4.4 Hadith Matn Verification

Hadith wording can vary more than Qur'anic text. For this reason, we combine lexical retrieval with character TF–IDF over character grams of length 3 to 7. The lexical matching threshold is 0.840. We may still accept a lower lexical score when its TF–IDF similarity reaches 0.90.

Very short *matn* segments are handled carefully because they may match many unrelated records. A one-token *matn* is rejected. A Qur'anic fragment placed in a Hadith *matn* field is also rejected as an incorrect segment type.

4.5 Claimed-Source Verification

For Qur'anic sources, the system extracts surah names, surah numbers, verse numbers, and verse ranges. It then compares the related Ayah with the canonical text at that exact location.

For Hadith sources, common Arabic names and spelling forms are mapped to the six books included in the task collection. The claimed book is checked against the records retrieved for the related *matn*. A source name alone is not enough; it must agree with the quotation.

4.6 Isnad Verification

We first remove common reporting phrases and honorific expressions from the *isnad*. we use the related *matn* to retrieve possible Hadith records. We then compare the narrator names only with the chain appearing before the Hadith text.

The system accepts the *isnad* when the best narrator-token coverage is at least 0.70. This approach is simple, but it does not handle spelling variations, long chains, or different forms of narrator names well.

Subtask	Metric	Test score
Subtask 1	Character macro-F1	0.9683
Subtask 2	Macro accuracy	0.9050

Table 2: Official blind-test results of the submitted systems.

Segment type	Accuracy (%)
Ayah	97.57
Claimed source	94.93
Matn	91.87
Isnad	77.63
Macro	90.50

Table 3: Subtask 2 accuracy by segment type on the blind test.

5 Results and Analysis

Table 2 reports the official blind-test results. The Subtask 1 system achieved 0.9683 character-level macro-F1. The Subtask 2 system achieved 0.9050 macro accuracy.

The high Subtask 1 score suggests that contextual BIO tagging and offset-based decoding can identify the different citation types with good boundary accuracy. Only the overall test score was provided, so We cannot provide a reliable per-type analysis for this subtask.

Table 3 gives the Subtask 2 results by segment type. Ayah verification achieved the highest accuracy at 97.57%. Claimed-source accuracy was 94.93%, while *matn* accuracy was 91.87%. These results suggest that retrieval from canonical texts works well for quotation and source checking.

The lowest accuracy is 77.63% for *isnad*. Narrator chains are more difficult because names may have different spellings, several narrators may appear in one chain, and the boundary between the chain and the Hadith text is not always clear. This result suggests that future work should use better Arabic name normalization and direct chain alignment.

6 Conclusion

We described Taqwatech's systems for the first two IslamicEval 2026 subtasks. For span detection, CAMeLBERT-Mix was fine-tuned with BIO labels and we converted its token predictions to character offsets. For hallucination identification, we used a source-based system that combined Arabic normalization, lexical retrieval, character TF–IDF, source checks, and narrator matching.

The submitted systems achieved 0.9683 character-level macro-F1 for Subtask 1 and 0.9050 macro accuracy for Subtask 2 on the official blind test. The results are strongest for span detection and Qur'anic verification. Improving the handling of Arabic narrator names and Hadith chains is the main direction for future work.

Limitations

The Qwen span-generation approach could not be evaluated in a directly comparable way and is not presented as a competitive result. The verification thresholds were selected using the released data and may not perform as well on other collections. The Hadith reference data covers six books and cannot confirm every source outside that collection. The *isnad* method uses token overlap and does not explicitly model complete narrator chains or narrator identity. Finally, only the overall Subtask 1 blind-test score is available, so we cannot give a reliable error analysis for each span type.

Ethical Considerations

The system is intended as a verification aid and not as a source of religious rulings. Text matching can find many quotation and source errors, but it cannot determine theological meaning or legal relevance. A false positive could accept fabricated content, while a false negative may reject an authentic wording variant. The predictions should therefore be checked against the retrieved evidence and reviewed by qualified experts when used in practice.

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